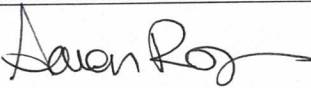


	Design Review		Doc No.: FM-0052
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from.</i>		Rev.: B
			Page 1 of 2

Project No.		Product Name	Part No./ Rev	Review Phase	Review Date
00017		Hydrophilix 100% Silicone Foley Catheter	HDX-1605 / A	Phase 5 Transfer to Manufacturing Manual Assembly	08/31/2020
Subject of Review					
#	Topic	Details		Reference Documents	
1	Action Items from Last Review (08/25/2020)	No action items		FILE-005991 rev 08/25/2020	
2	Market Clearance	FDA Letter SE 04/21/2020 RE: K192034. The requirement is complete.		K192034 rev 04/19/2020	
3	Device Master Record	The documents were reviewed. The requirement is complete.		HDX-1605 rev A	
4	Routine Sterilization Program	Hydrophilix products were adopted into Pathway's validated cycle no. PWI-7374. This file is an attachment QP-032 rev C EO Sterilization Process Control SOP. The requirement is complete.		QP-032 rev C	
5	Supplier Qualification	Existing suppliers Steripax and Steris are on the Approved Supplier List. They were last assessed on ECO-000415, ECO-000534. NingoYingmed was approved as a new supplier per ECO-000488. The requirement is complete.		ECO-000415, ECO-000488, ECO-000534.	
6	Process FMEA	The document was reviewed. The requirement is complete.		RM-0021 rev A	
7	Process Validation	Process validation is not required for 100% verifiable processes. All of the processes with the exception of pouch sealing, is 100% verifiable. Pouch sealing process has been validated per TR-SAL-000, TR-SAL-0003, and TR-SAL-0004. Therefore, for the manual process, the requirement is complete.		N/A	
8	Open Items	The process validation for the automated surface treatment process is outstanding.		N/A	
9	Project Plan	No changes required to the project plan. The plan is current.		PL-0013 rev B	
10	Risk	No new risks discovered, nor changes required to the RMP or HA or DFMEA. The risk management plan is current. With no new risks identified.		RM-0018 rev A, RM-0019 rev A, RM-0020 rev A	
11	Review of Phase 5 Deliverables	Each document was reviewed and discussed. Project team in attendance has concluded that the deliverables are complete and satisfy the Design Control requirements in accordance with QP-007 rev E. The requirements for Transfer to Manufacturing have been fulfilled for the manual surface treatment process. The project team agrees to release first lot for commercial distribution under limited market release.		Refer to #1-#10, QP-007 rev E.	
Action Items					
#	Description		Assigned To		Due Date
N/A	N/A		N/A		N/A

	Design Review	Doc No.: FM-0052
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from.</i>	Rev.: B
		Page 2 of 2

Project No.	Product Name	Part No./ Rev	Review Phase	Review Date	
00017	Hydrophilix 100% Silicone Foley Catheter	HDX-1605 / A	Phase 5 Transfer to Manufacturing Manual Assembly	08/31/2020	
Reviewers					
Functional Group	Required?	Name	Title	Signature	Date
Quality Assurance / Regulatory Affairs	☒	Aaron Rogers	Director of Regulatory and Quality		09/01/2020
R&D/ Engineering / Manufacturing	☒	Jonas Cochran	Managing Partner		9/01/2020
Marketing	☐				
Independent Reviewer	☒	Tom Cassou	Director of Business Development		9/1/2020
Other / Customer	☒	Brian McVerry	Chief Technology Officer		9.1.2020



April 21, 2020

Pathway, LLC
Aaron Rogers
Director of Regulatory and Quality
8779 Cottonwood Avenue, Suite 105
Santee, CA 92071

Re: K192034
Trade/Device Name: HDX 100% Silicone 2-way Foley Catheter, 14Fr/10cc
HDX 100% Silicone 2-way Foley Catheter, 16Fr/5cc
HDX 100% Silicone 2-way Foley Catheter, 18Fr/10cc
Regulation Number: 21 CFR 876.5130
Regulation Name: Urological catheter and accessories
Regulatory Class: II
Product Code: EZL
Dated: March 19, 2020
Received: March 23, 2020

Dear Aaron Rogers:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database located at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling

(21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for devices or postmarketing safety reporting (21 CFR 4, Subpart B) for combination products (see <https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance>) and CDRH Learn (<https://www.fda.gov/training-and-continuing-education/cdrh-learn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,


Purva U. Pandya -S

Purva Pandya

Acting Assistant Director

DHT3B: Division of Reproductive,

Gynecology and Urology Devices

OHT3: Office of GastroRenal, ObGyn,

General Hospital and Urology Devices

Office of Product Evaluation and Quality

Center for Devices and Radiological Health

Enclosure



510(K) SUMMARY – K192034

Submitter Information

Company Name: Hydrophilix, Inc.
Company Address: 12100 Wilshire Blvd
Suite 800
Los Angeles, CA 90025

Contact Person: Aaron Rogers
Pathway, LLC.
Director of Regulatory & Quality
(619) 415-0103
arogers@pathwaympi.com

Date: April 21, 2020

Device Identification

Device Trade Name: HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 14Fr/10cc
HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 16Fr/5cc
HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 18Fr/10cc

Common Name: Foley Catheter

Classification Name: Urological catheter and accessories

Classification Number: 876.5130

Regulatory Class: Class II

Product Code(s): EZL

Product Code Name: Catheter, Retention Type, Balloon

Advisory Panel: Gastroenterology/Urology

Identification of Predicate Device

Predicate Device				
Device Name	Regulation No.	Product Code	510(K) Number	Clearance Date
Bardex Lubi-Sil Foley Catheter	876.5130 - Urological catheter and accessories	EZL	K984084	February 1, 1999

Device Description

The Subject Device is a single-use, 2-way Foley catheter that is constructed of medical grade silicone. It incorporates two (2) lumens, one for inflation/deflation of the balloon and the other for drainage of the urinary tract.

The balloon, at the distal end of the catheter, is inflated to the labeled fill volume and remains inflated via the incorporated two-way valve in the connector at the proximal end of the inflation lumen. The connector engages with slip/taper luer connection that is standard on syringes. With the balloon inflated, the catheter maintains position in the urinary tract for the duration of the clinical application. Prior to removal, the balloon is deflated via the 2-way valve connection.

The drainage inlet is located distal to the catheter's balloon. The connection to the urinary bag is a standard non-interconnectable connector.

The Subject Device is packaged inside a perforated plastic sleeve that is inserted into a sterile barrier pouch. Ten (10) units are packaged into a carton, which is the saleable unit. The Subject Device is terminally sterilized via Ethylene Oxide (EO).

The Subject Device will be initially offered in 14, 16, 18Fr sizes. The entire length of the Subject Device is disposable. The Subject Device is not sold as a set, there are no accessories or components included.

Indications for Use

Urological catheter intended for drainage of the urinary tract. Catheterization is accomplished by inserting the catheter through the urethra and into the bladder. Intended population is adults and pediatrics.

Comparison of Technological Characteristics with Predicate Device

Comparison Table – Subject and Predicate Devices		
Comparison Feature	Subject Device	Predicate Device
Indications for Use	Urological catheter intended for drainage of the urinary tract. Catheterization is accomplished through the urinary tract. Intended population is adults and pediatrics.	Used in the drainage and/or collection and/or measurement of urine. Generally, drainage is accomplished by inserting the catheter through the urethra and into the bladder. However, drainage is sometimes accomplished by suprapubic or other placement of the catheter, such as a nephrostomy tract.
Type	2-way with inflation and drainage lumens	2-way with inflation and drainage lumens
Size/Balloon Size	14Fr/10cc 16Fr/5cc 18Fr/10cc	8, 10Fr/3cc Balloon 12-24Fr, even sizes/5cc Balloon 16-24Fr, even sizes/30cc Balloon
Material	Silicone, Medical Grade	Silicone, High Grade
Performance Standard	ASTM F623	ASTM F623
Single-Use?	Yes	Yes
Prescription Use?	Yes	Yes
Sterile?	Yes	Yes

Summary of Evaluations Performed

The test program was performed in accordance to FDA guidance and recognized performance states, which includes the following:

- Biocompatibility
- Sterilization
- Packaging Integrity (i.e., Sterile Barrier)

- Transportation
- Performance/Functionality

Successful results were achieved with all evaluations conducted.

Conclusion

The Subject Device has demonstrated it is substantially equivalent to the commercially available predicate.

Indications for Use

510(k) Number (if known)

K192034

Device Name

HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 14Fr/10cc

HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 16Fr/5cc

HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 18Fr/10cc

Indications for Use (Describe)

Urological catheter is intended for drainage of the urinary tract. Catheterization is accomplished by inserting the catheter through the urethra and into the bladder. Intended population is adults and pediatrics.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

This section applies only to requirements of the Paperwork Reduction Act of 1995.

DO NOT SEND YOUR COMPLETED FORM TO THE PRA STAFF EMAIL ADDRESS BELOW.

The burden time for this collection of information is estimated to average 79 hours per response, including the time to review instructions, search existing data sources, gather and maintain the data needed and complete and review the collection of information. Send comments regarding this burden estimate or any other aspect of this information collection, including suggestions for reducing this burden, to:

Department of Health and Human Services
Food and Drug Administration
Office of Chief Information Officer
Paperwork Reduction Act (PRA) Staff
PRAStaff@fda.hhs.gov

"An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB number."



Pathway LLC

Printed by Aaron Rogers

Printed on August 25, 2020

Local time zone (GMT-08:00) Pacific Time (America/Los_Angeles)

Page Arena > Workspace Items > Bill of Materials > Indented

All information contained in this document is proprietary and confidential.

HDX-1605

In Production

Hydrophilix 16 Fr x 5 ml Catheter, Sterile, 10-Pack

Revision »A - ECO-000652 **Effective** as of 08/24/2020 10:35:00 AM, Unshared

Contains **4** first-level Items, **28** line Items, **27** unique Items, **1** of which is shared.

▼ # ITEM NUMBER	ITEM NAME	CATEGORY	PHASE	WKG MODS	WHERE USED	FILES	RQMTS	QTY	BOM NOTES
1 FM-0029 rev A	Form, Certificate of Conformance	Form	In Prod				2	1 each	
▼ 2 HDX-8001 rev B	Hydrophilix 16 Fr x 5ml Catheter, Non-Sterile, 10-Pack	Final Assembly	In Prod				3	1 EA	
1 MP-HDX-0000 rev A	Manufacturing Procedure, Hydrophilix Polymer Coating Application Utilizing UV Reactor	Manufacturing Procedure	In Prod				3	1 EA	
2 MP-HDX-0001 rev A	Manufacturing Procedure, Labeling and Packaging, Hydrophilix Polymer Coated Catheters	Manufacturing Procedure	In Prod				3	1 EA	
3 IP-0002 rev B	Inspection Procedure, Product Labeling and Packaging	Inspection Procedure	In Prod				1	1 each	
4 MP-0016 rev A	Manufacturing Procedure, Label Generation	Manufacturing Procedure	In Prod				2	1 each	Max 12 month product shelf life
5 HDX-6001 rev A	Catheter, 16 Fr x 5 ml, Silicone, NY Foley	Purchased Component	In Prod				7	10 EA	
6 HDX-3000 rev A	Solution, Hydrophilix Polymer	Purchased Component	In Prod				7	80 ML	
7 HDX-5000 rev A	Bag, 2 W x 18 L x 2 Mil Thk, Polyethylene	Packaging	In Prod				3	10 EA	No Lot Number is Required
8 HDX-4000 rev C	Artwork, Label, Hydrophilix Catheter, Pouch and Box	Labeling	In Prod				1	12.2 EA	No Lot Number is Required
9 HDX-4001 rev C	Artwork, Label, Hydrophilix Catheter, IFU	Labeling	In Prod				2	10 EA	
10 HDX-5001 rev A	Pouch, Chevron, 3.5 X 21, Amcor TLP-058	Packaging	In Prod				3	10 EA	
11 HDX-5002 rev A	Box, 10-Pack, Inner, 20.75 L x 4.12 W x 2.37 H, PSM	Packaging	In Prod				2	1 EA	No Lot Number is Required

▼ # ITEM NUMBER	ITEM NAME	CATEGORY	PHASE	WKG WHERE		FILES	RQMTS	QTY	BOM NOTES
				MODS	USED				
12 HDX-5003 rev A	Box, 22 in. L x 14 in. W x 12 in. H, 275 LB Test, Double Wall	Packaging	In Prod				2	0.1 EA	No Lot Number is Required
13 HDX-5004 rev B	Label, 2.3125 in. X 7.50 in.	Packaging	In Prod				2	12.2 EA	No Lot Number is Required
14 HDX-5005 rev A	Ribbon, Thermal Transfer, Resin/Wax, 4.3" W	Packaging	In Prod				1	92 IN	No Lot Number is Required
15 HDX-5006 rev A	Tape, Packaging, 2 Mil, 2" x 110 yds, Clear	Packaging	In Prod				2	560 IN	No Lot Number is Required
▼ 3 QP-010 rev F	Final Product Inspection	Quality System Procedure	In Prod				6	1 each	
1 FM-0029 rev A	Form, Certificate of Conformance	Form	In Prod				2	1 each	
2 FM-0054 rev A	Packing List Log	Form	In Prod				1	1 each	
3 FM-0062 rev A	Form, Packing List	Form	In Prod				2	1 each	
4 T-002 rev A	Template, Inspection Record	Template	In Prod				2	1 each	
5 WI-0021 rev A	Work Instruction, Pyrogen (LAL) Evaluation Process	Work Instruction	In Prod				2	1 each	
▼ 4 QP-032 rev C	EO Sterilization Process Control	Quality System Procedure	In Prod				4	1 each	
1 21 CFR 801 rev 04/01/2019	Labeling	External Document	In Prod				1	1 each	
2 ANSI/AAMI/ISO-10993-7 rev 2008(R)2012	Biological evaluation of medical devices— Part 7: Ethylene oxide sterilization residuals	External Document	In Prod				1	1 each	
3 ANSI/AAMI/ISO-11135 rev 2014	Sterilization of health care products— Ethylene oxide — Requirements for development, validation	External Document	In Prod				1	1 each	
4 FM-0031 rev A	Form, Pallet Plaque, Non-Sterile (For Further Processing)	Form	In Prod				2	1 each	

	Process Failure Mode Effects Analysis (FMEA) – Risk Management Report Hydrophilix Foley Catheter		Doc No.: RM-0021
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>		Rev.: A
			Page 1 of 7

1.0 Purpose

- 1.1 The purpose of this document is to identify the associated risk with the manufacturing processes of the Hydrophilix Foley Catheter.

2.0 Scope

- 2.1 The product numbers associated with the Hydrophilix Foley Catheter is indicated in the following table:

Table 2.1 – Hydrophilix Foley Catheter Products	
Part No.	Description
HDX-1410	HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 14Fr/10cc
HDX-1605	HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 16Fr/5cc
HDX-1810	HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 18Fr/10cc

3.0 References

- 3.1 QP-004 – Risk Management

4.0 Risk Tables Definitions

- 4.1 # – Enumerate each Description with a sequential number
- 4.2 Description – Describes the operation that is being evaluated for risk
- 4.3 Potential Failure Mode (FM) – The potential consequences of the failure
- 4.4 Severity (S) – Measure of the possible consequences of a hazard.
- 4.5 Potential Effect of Failure – All conceivable effects of the potential failure mode
- 4.6 Potential Cause of Failure – All conceivable causes of the potential failure mode
- 4.7 Risk Controls - Process through which decisions are reached and protective measures are implemented for reducing risk to, or maintaining risk within specified levels.
- 4.7.1 Risk Controls¹ – Initial risk controls that are incorporated.
- 4.7.2 Risk Controls² – Secondary risk controls, also known as “Mitigation”.
- 4.8 Verification – Evidence source of linkage that initial (i.e., Risk Controls¹) risk controls have implemented, verified, or validated.
- 4.9 Probability – Likelihood of the hazard occurring

	Process Failure Mode Effects Analysis (FMEA) – Risk Management Report Hydrophilix Foley Catheter		Doc No.: RM-0021
	This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work.		Rev.: A
			Page 2 of 7

4.9.1 Probability¹ (P¹) – The probability assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)

4.9.2 Probability² (P²) – The probability assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)

4.10 Detection - The ability to screen out a hazard through verification activities (e.g., inspection).

4.10.1 Detection¹ (D¹) – The detection assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)

4.10.2 Detection² (D²) – The detection assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)

4.11 Frequency – A 'Frequency Score' will be calculated using the ratings from probability and detection components. The frequency equation is the following: ***Frequency = $\sqrt{Px\bar{D}}$*** **Note: The value calculated should be rounded up.**

4.11.1 Frequency¹ (F¹) – The frequency score calculated using Risk Controls¹ and Detection¹

4.11.2 Frequency² (F²) – The frequency score calculated using Risk Controls² and Detection²

4.12 Risk Rating – Residual risk determination based on the concept of 'As Far As Possible' (AFAP) approach. The AFAP approach provides three (3) possible outcomes, which are 1) the residual risk exceeds the risk acceptability (RED), 2) the residual risk is acceptable to the risk acceptability criteria (GREEN), and 3) the residual risk is between the two (2) states indicated and it can be an option to lower the risk to the lowest practical level without economic considerations (YELLOW).

4.12.1 Risk Rating¹ (R¹) – This calculation is post the implementation of initial risk controls (i.e., Risk Controls¹)

4.12.2 Risk Rating² (R²) – This calculation is post the implementation of the secondary risk controls (i.e., Risk Controls²)

5.0 Risk Acceptability Determination

5.1 The criteria of risk acceptability will be determined with a Risk Priority Number (RPN) and on the concept of 'As Far As Possible (AFAP)' approach. Refer to the Risk Rating definition in §4.0.

5.2 There are three (3) components used in determining risk acceptability, which are Severity, Probability, and Detection. Refer to the following tables:

Table 5.2a - Severity Ratings	
Index	Description
1	No patient effect
2	Minor injury that resolves without treatment
3	Moderate injury that may require treatment
4	Serious injury requiring medical intervention
5	Life threatening injury, disability, death

Table 5.2b – Probability Ratings	
Index	Description

	Process Failure Mode Effects Analysis (FMEA) – Risk Management Report Hydrophilix Foley Catheter											Doc No.: RM-0021
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>											Rev.: A
												Page 3 of 7

Table 5.2b – Probability Ratings	
Index	Description
1	Improbable - <0.001%
2	Remote – 0.001% to .01%
3	Occasional – 0.01% to 0.1%
4	Likely – 0.1% to 1%
5	Frequent - >1%

Table 5.2c – Detection Ratings	
Index	Description
1	Very High – Current controls are almost certain to detect the failure mode.
2	High – High likelihood that current controls will detect the failure mode.
3	Moderate – Moderate likelihood that current controls will detect the failure mode.
4	Low – Low likelihood that current controls will detect the failure mode.
5	Remote – Remote likelihood that current controls will detect the failure mode.

5.3 Risk acceptability will be determined per the following risk index:

Table 5.3 – Risk Rating Table						
Severity	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
	1	1	2	3	4	5

=Square Root (Probability X Detection)

Note: The purpose of using the square root of the probability and detection scores is to deemphasize these two (2) components, which places more emphasize on the severity component.

6.0 Risk Assessment Tables

Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
1	Receiving Inspection	Accept Non-Conforming Material	2	Procedural Complication	Operator Error	Process Instructions	QP-009	1	1	1	2	N/A	N/A	N/A	N/A	N/A
			1		Equipment Error	Calibration Program	QP-020	1	1	1	1	N/A	N/A	N/A	N/A	N/A

	Process Failure Mode Effects Analysis (FMEA) – Risk Management Report Hydrophilix Foley Catheter											Doc No.: RM-0021			
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>											Rev.: A			
												Page 4 of 7			

Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
2	Line Clearance / Area Preparation	Incorrect Material	1	Non-Functional Device	Operator Error	Process Instructions Process Settings/Line Clearance Form DMR	WI-0014 FM-0039	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Material Expired	1	Non-Functional Device	Operator Error	DMR	FM-0039	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Improper Material Handling	1	Damaged Materials	Operator Error	Process Instructions	WI-0014	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Out of Calibration Equipment	3	Non-Functional Device	Operator Error	Process Instructions Equipment Qualification	WI-0014 IQ Report (TR)	1	1	1	3	N/A	N/A	N/A	N/A	N/A
		Untrained Operators	1	Compromised Device	Inadequate Training	Process Instructions Training Program	WI-0014 QP-0017	1	1	1	1	N/A	N/A	N/A	N/A	N/A
3	Catheter tube forming / curing	Incomplete / Malformed Tube	2	Non-Functional Device / Poor flow	Operator Error	Supplier Qualification, Receiving Inspection In-Process Inspection	Ningbo Yingmed Catheter Inspection Plans (Form) HDX-6XXX	1	2	2	4	N/A	N/A	N/A	N/A	N/A

	Process Failure Mode Effects Analysis (FMEA) – Risk Management Report Hydrophilix Foley Catheter											Doc No.: RM-0021			
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>											Rev.: A			
												Page 5 of 7			

Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
4	Balloon forming / dipping	Incomplete / Malformed Balloon	3	Non-Functional Device / Balloon Burst	Operator Error	Supplier Qualification, Receiving Inspection In-Process Inspection	Ningbo Yingmed Catheter Inspection Plans (Form) HDX-6XXX	1	2	2	6	N/A	N/A	N/A	N/A	N/A
5	Valve installation	Loose valve / Valve disconnects	1	Non-Functional Device	Operator error	Supplier Qualification, Receiving Inspection In-Process Inspection	Ningbo Yingmed Catheter Inspection Plans (Form) HDX-6XXX	1	1	1	1	N/A	N/A	N/A	N/A	N/A
6	Labeling	Incorrect print information	1	Potential traceability issue	Operator error	Process instruction	MP-0016	2	1	2	2	Label inspection per IP-0002	1	1	1	1
		Illegible	1	Potential traceability issue	Equipment error	Process instruction	MP-0016	2	1	2	2	Label inspection per IP-0002	1	1	1	1
7	Surface Treatment	Incomplete / No Treatment	1	No effect	Equipment error	Process instruction	MP-HDX-0000	1	1	1	1	N/A	N/A	N/A	N/A	N/A
						In-Process Inspection	Inspection Plans (Form) HDX-8XXX									
					Operator error	Process instruction	MP-HDX-0000									
						In-Process Inspection	Inspection Plans (Form) HDX-8XXX									
8	Apply Label to Pouch	Misaligned or wrinkled	1	Cosmetic issue	Operator Error	Process Instructions	MP-HDX-0001	1	1	1	1	N/A	N/A	N/A	N/A	N/A
9	Pouch Loading	Incorrect product configuration	1	Non-Functional Device	Operator error	Process instruction	MP-HDX-0001	2	2	2	4	N/A	N/A	N/A	N/A	N/A

	Process Failure Mode Effects Analysis (FMEA) – Risk Management Report Hydrophilix Foley Catheter											Doc No.: RM-0021
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>											Rev.: A
												Page 6 of 7

Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
10	Tray Sealing (Sterile Barrier)	Weak Seal	4	Breach in sterile barrier	Equipment error	Process validation Inspection process	TR-SAL-0000 (IQ), TR-SAL-0003 (OQ), TR-SAL-0004 (PQ) Inspection Plans (Form) HDX-8XXX	1	1	1	4	N/A	N/A	N/A	N/A	N/A
					Operator error – incorrect sealing parameters	Process instruction Inspection process	TR-0074 Inspection Plans (Form) HDX-8XXX	1	1	1	4	N/A	N/A	N/A	N/A	N/A
					Contamination	Environmental controls	QP-018	1	1	1	4	N/A	N/A	N/A	N/A	N/A
11	Assemble Box	Incorrect / Weak box	1	Damage product	Operator error	Process instruction	MP-HDX-0001	1	1	1	1	N/A	N/A	N/A	N/A	N/A
	Apply Label to Box	Misaligned or wrinkled	1	Cosmetic issue	Operator Error	Process Instructions	MP-HDX-0001	1	1	1	1	N/A	N/A	N/A	N/A	N/A
12	Packaging	Incorrect packaging configuration	2	Damage product	Operator error	Process instruction	MP-HDX-0001	2	2	2	4	Packaging inspection per IP-0002	1	1	1	2
13	Final Product Inspection	Incomplete DHR	1	Potential traceability issue	Operator error	Process instruction	WI-0006 QP-010	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Incorrect DHR	1	Potential traceability issue	Operator error	Process instruction	WI-0006 QP-010	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		No CofC	1	N/A	Operator error	Process instruction	QP-010	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Incomplete CofC	1	N/A	Operator error	Process instruction	QP-010	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Incorrect CofC	1	N/A	Operator error	Process instruction	QP-010	1	1	1	1	N/A	N/A	N/A	N/A	N/A

	Process Failure Mode Effects Analysis (FMEA) – Risk Management Report Hydrophilix Foley Catheter							Doc No.: RM-0021
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>							Rev.: A
								Page 7 of 7

7.0 Risk/ Benefit Assessment

7.1 The following table indicates the risk/benefit analysis for the residual risks that were greater than the risk acceptability limit after all applicable controls/mitigations were applied:

Table 7.1 – Risk/Benefit Assessment for Residual Risk above Acceptability Limit for Process Hazard Assessment									
Table No.	Process Description	Section No.	Potential Failure Mode	Potential Effect of Failure	Potential Cause of Failure	S	F ²	Risk/Benefit Analysis	Does the benefit outweigh the residual risk?
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

7.2 There was no residual risk identified that was above the risk acceptability limit per §5.3.

8.0 Conclusion

8.1 Risks have been reduced as low as possible by the following methods:

- Design the product with inherent safety
- Appropriate testing to ensure design criteria were met
- Protective measures/ mechanisms to the product of the device. (critical-to-quality)
- Manufacturing and quality control measures
- Provide information for safety

Based on a review of the risk-management process for the Hydrophilix Foley Catheter described herein, the risk management plan based on EN ISO 14971:2012 has been implemented and the individual, cumulative and overall residual risk have been reduced as low as possible regardless of the magnitude and the benefit of using this device will outweigh risks posed by the device during usage.

9.0 Change History

Revision	Change Description
A	Initial release

	Design Development Plan, Hydrophilix Foley Catheter		Doc No.: PL-0013
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work</i>		Rev.:B
			Page 1 of 3

1.0 Purpose

- 1.1 The purpose of this document is to provide details associated with design and development project for initial market release of the Hydrophilix (HDX) Foley Catheter. This project is documented under PRJ-0017.

2.0 Scope

- 2.1 The plan applies to the HDX Foley Catheter described below, which will be initial distributed in the US.

3.0 References

- 3.1 PRJ-0017 – Design Project, Hydrophilix Foley Catheter Product Introduction (US)
- 3.2 QP-007– Design Control

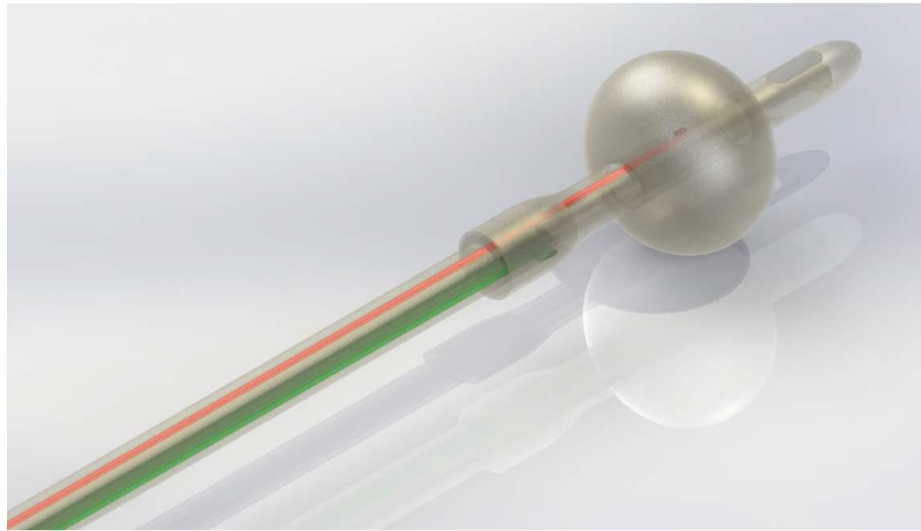
4.0 Background

- 4.1 Hydrophilix, Inc. was spun-out of the University of California, Los Angeles based on the Hydrophilix coating technology, a direct result of federal, state, and local support. The proof-of-concept began with basic research funding from the National Science Foundation Sustainable Chemistry Program and fellowships for individual scholars. The technology was further developed for commercial application, propelled with seed funding from the NSF Clean Energy for Green Industry, UCLA's Physical Sciences Entrepreneurship and Innovation Fund, and private investment.
- 4.2 When devices are processed with the Hydrophilix solution, a thin hydration layer prevents pathogens from adhering to the device surface and proliferating into biofilms. Because the processed surface does not kill microbes, the creation of superbugs is avoided. The processed surface is non-toxic and is applied in minutes.

5.0 Product Description

- 5.1 The HDX Foley catheter is a single use device that intended for use in the drainage and/or collection and/or measurement of urine. It is constructed of 2 lumens, which are for balloon inflation and drainage. The catheter will be constructed of medical grade silicone. The size offering will be 14, 16, and 18Fr with balloon volumes of 5ml – 30 ml. Refer to the following illustration of the device:

	Design Development Plan, Hydrophilix Foley Catheter		Doc No.: PL-0013
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work</i>		Rev.:B
			Page 2 of 3



6.0 Product Configuration

- 6.1 The product will be offered in 3 sizes initially: 14Fr 10ml, 16Fr 5ml, and 18Fr 10ml. Additional sizes will be added as they are available.
- 6.2 Devices will be packaged individually in a sterile barrier Tyvek/poly pouch, which 10 units will be incorporated per box.

7.0 Indication for Use Statement

- 7.1 Urological catheter intended for drainage/irrigation of urinary tract. Catheterization is accomplished by inserting the catheter through the urethra and into the bladder. Intended population is adults and pediatrics.

8.0 Goals & Objectives

- 8.1 Complete the following Design Control Activities defined in QP-007
 - 8.1.1 Planning
 - 8.1.2 Design Input
 - 8.1.3 Design Output
 - 8.1.4 Design Verification and Validation
 - 8.1.5 Prepare and submit FDA pre-market notification (510k) application and gain clearance to distribute in the US.
 - 8.1.6 Transfer the product to manufacturing to support commercialization of product. Note: This phase will be split into the following two (2) subphases.
 - 8.1.6.1 Manual process
 - 8.1.6.2 Automated process

9.0 Resources

	Design Development Plan, Hydrophilix Foley Catheter		Doc No.: PL-0013
	<i>This document contains confidential, proprietary information of Pathway LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled "For Reference Only", unless accompanied with a controlled stamp. Users of this document must verify that they are using the most current revision of the controlled document before performing work</i>		Rev.:B
			Page 3 of 3

- 9.1 Refer to the following table for the internal and external resources that are required to execute the design and development plan:

Internal/ External	Function	Name	Title
External	Customer	Brian McVerry	CTO
Internal	R&D	Arthur Deptala	Managing Partner
Internal	Manufacturing / Project Leader	Jonas Cochran	Managing Partner
Internal	Regulatory/ Quality	Aaron Rogers	Director of Regulatory and Quality
Internal	Independent Reviewer	Tom Cassou	Director of Business Development

- 9.2 External testing services are provided by:

9.2.1 Biocompatibility: Nelson Labs, WUXI

9.2.2 Sterilization: Steris

9.2.3 Transportation Testing: Westpak

- 9.3 Key Component and Process Suppliers:

9.3.1 Non-Sterile Foley Catheter: Ningbo Yingmed

9.3.2 Surface Treatment Solution: Hydrophilix, Inc.

9.3.3 Sterile Barrier Packaging: Steripax

9.3.4 Sterilization: Steris

10.0 Schedule/ Timelines

- 10.1 Schedules and timelines for each phase will be updated throughout the project and is maintained as part of the project module in Arena PLM.

11.0 Change History

Revision	Change Description
A	Initial Release
B	Update resources §9.0, Indications for Use §7.0, project phases §8.0, product configuration §6.0

Pathway Innovation LLC.	Risk Management Plan – Hydrophilix Foley Catheter	Doc No.: RM-0018
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from.</i>	Rev.: A
		Page 1 of 2

1.0 Purpose

The purpose of this document is to indicate the requirements of the risk management plan that is associated with the development of the Hydrophilix (HDX) Foley Catheter.

2.0 Scope

2.1 The document applies for the design project associated with the product introduction of the HDX Foley Catheter.

2.2 This document will identify the activities planned for risk management.

3.0 References

3.1 ISO 14971:2012 – Medical devices – Application of risk management to medical devices

3.2 QP-002 – Document Control

3.3 QP-004 – Risk Management

3.4 QP-007 – Design Control

3.5 QP-011 – Post Market Surveillance

3.6 Project#00017 – Hydrophilix Foley Catheter product introduction (US)

4.0 Responsibilities & Authorities

4.1 The risk management team will include members of Pathway. Team members will include individuals representing Research & Development (R&D), Manufacturing, Marketing, Regulatory Affairs (RA), and Quality Assurance (QA).

Individuals that participate in risk management activities will have knowledge of and experience with the device (or similar device) and how it is manufactured.

4.2 The risk management team will be required to complete all activities indicated in this document. Team members from Pathway will be required to review all content of risk management deliverables and agree on its conclusions with documented approval.

5.0 Product Description

The HDX Foley Catheter is intended for use to drain/irrigate the urinary tract.

6.0 Product Life-Cycle Phases

6.1 Design

In the appropriate phase of the design control process, per QP-007, a Hazard Analysis (HA) and Design Failure Mode Effects Analysis (DFMEA) will be developed per, QP-004, as input to the design of the device.

Criteria of risk acceptability is identified in QP-004.

Pathway Innovation LLC.	Risk Management Plan – Hydrophilix Foley Catheter	Doc No.: RM-0018
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from.</i>	Rev.: A
		Page 2 of 2

If additional risk(s) are identified during the progression through the design control phases, especially in verification and validations, risk management deliverables will be reviewed and updates appropriately.

6.2 Process Validation

In the “Transfer to Manufacturing” phase of the design control process per QP-007, a Process Failure Mode Effects Analysis (PFMEA) will be developed as a deliverable process validation. Criteria of risk acceptability is identified in QP-004.

6.3 Production & Post-Production

Information collected from production and post-production activities will be evaluated for risk as part of the post-market surveillance process per QP-011. Risk management report will be review and newly identified risk or changes to existing risk will be incorporated.

Revised information to the risk management reports will be reviewed with R&D, RA, and QA to determine if action shall be taken to the product design to mitigate identified risk. Action shall be incorporated in a change control project per QP-007.

7.0 Verification Activities

Activities associated for risk controls (initial & mitigation) will be documented in the risk management reports.

8.0 Risk Management File

The risk management file consists of this plan and the risk management report (i.e., Hazard Analysis, DFMEA, & PFMEA). The risk management plan will be maintained as a controlled document per QP-002 and associated in the product’s Design History File (DHF).

9.0 Change History

Revision	Change Description
A	Initial release

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter	Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 1 of 9

1.0 Purpose

- 1.1 The purpose of this document is to identify the associated risk with the intended use of the HDX Foley Catheter.

2.0 Scope

- 2.1 The product numbers associated with the HDX Foley Catheter is indicated in the following table:

Table 2.1 – HDX Foley Catheter Products	
Part No.	Description
HDX-8000	14 Fr, 2-Way Foley Catheter
HDX-8001	16 Fr, 2-Way Foley Catheter
HDX-8002	18 Fr, 2-Way Foley Catheter

- 2.2 The intended use of this device is the following:

Single-use urinary catheter that intended for use in the drainage / irrigation of the urinary tract.

3.0 References

- 3.1 QP-004 – Risk Management

4.0 Risk Tables Definitions

- 4.1 Description – Describes the operation that is being evaluated for risk
- 4.2 Verification – Evidence source of linkage that initial (i.e., Risk Controls¹) risk controls have implemented, verified, or validated.
- 4.3 Probability – Likelihood of the hazard occurring
- 4.3.1 Probability¹ (P¹) – The probability assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)
- 4.3.2 Probability² (P²) – The probability assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)
- 4.4 Potential Hazard (PH) – Source of harm
- 4.5 Potential Failure Mode (FM) – The potential consequences of the failure
- 4.6 Risk Controls - Process through which decisions are reached and protective measures are implemented for reducing risk to, or maintaining risk within specified levels.
- 4.6.1 Risk Controls¹ – Initial risk controls that are incorporated.
- 4.6.2 Risk Controls² – Secondary risk controls, also known as “Mitigation”.

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter	Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 2 of 9

4.7 Risk Rating – Residual risk determination based on the concept of ‘As Far As Possible (AFAP)’ approach. The AFAP approach provides three (3) possible outcomes, which are 1) the residual risk exceeds the risk acceptability (RED), 2) the residual risk is acceptable to the risk acceptability criteria (GREEN), and 3) the residual risk is between the two (2) states indicated and additional controls are suggested to lower the risk to the lowest practical level, without economically considerations, based on management decision (YELLOW).

4.7.1 Risk Rating¹ (R¹) – This calculation is post the implementation of initial risk controls (i.e., Risk Controls¹)

4.7.2 Risk Rating² (R²) – This calculation is post the implementation of the secondary risk controls (i.e., Risk Controls²)

4.8 Severity (S) – Measure of the possible consequences of a hazard.

5.0 Identify Characteristics that could Impact on Safety (Annex C)

C.2.1 What is the intended use and how is the medical device to be used? *Refer to §2.2.*

C.2.2 Is the medical device intended to be implanted? *No*

C.2.3 Is the medical device intended to be in contact with the patient or other persons? *Yes*

C.2.4 What materials or components are utilized in the medical device or are used with, or are in contact with, the medical device? *Silicone*

C.2.5 Is energy delivered to or extracted from the patient? *No*

C.2.6 Are substances delivered to or extracted from the patient? *Yes, substances can be delivery to the urinary tract.*

C.2.7 Are biological materials processed by the medical device for subsequent re-use, transfusion or transplantation? *No*

C.2.8 Is the medical device supplied sterile or intended to be sterilized by the user, or are other microbiological controls applicable? *Yes*

C.2.9 Is the medical device intended to be routinely cleaned and disinfected by the user? *No*

C.2.10 Is the medical device intended to modify the patient environment? *No*

C.2.11 Are measurements taken? *Yes, measurements of urine.*

C.2.12 Is the medical device interpretative? *No*

C.2.13 Is the medical device intended for use in conjunction with other medical devices, medicines or other medical technologies? *Yes, urine collection bag.*

C.2.14 Are there unwanted outputs of energy or substances? *No*

C.2.15 Is the medical device susceptible to environmental influences? *No*

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter	Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 3 of 9

- C.2.16 Does the medical device influence the environment? *No*
- C.2.17 Are there essential consumables or accessories associated with the medical device? *No*
- C.2.18 Is maintenance or calibration necessary? *No*
- C.2.19 Does the medical device contain software? *No*
- C.2.20 Does the medical device have a restricted shelf-life? *Yes*
- C.2.21 Are there any delayed or long-term use effects? *No*
- C.2.22 To what mechanical forces will the medical device be subjected? *Tensile force from the urine content in the collection bag; Internal pressure from urine drainage or irrigation of the urinary tract.*
- C.2.23 What determines the lifetime of the medical device? *Product and sterile barrier performance*
- C.2.24 Is the medical device intended for single use? *Yes*
- C.2.25 Is safe decommissioning or disposal of the medical device necessary? *Yes*
- C.2.26 Does installation or use of the medical device require special training or special skills? *No*
- C.2.27 How will information for safe use be provided? *Direction for Use*
- C.2.28 Will new manufacturing processes need to be established or introduced? *No*
- C.2.29 Is successful application of the medical device critically dependent on human factors such as the user interface? *No*
 - C.2.29.1 Can the user interface design features contribute to use error? *No*
 - C.2.29.2 Is the medical device used in an environment where distractions can cause use error? *No*
 - C.2.29.3 Does the medical device have connecting parts or accessories? *Yes, urine collection bag*
 - C.2.29.4 Does the medical device have a control interface? *No*
 - C.2.29.5 Does the medical device display information? *No*
 - C.2.29.6 Is the medical device controlled by a menu? *No*
 - C.2.29.7 Will the medical device be used by persons with special needs? *No*
 - C.2.29.8 Can the user interface be used to initiate user actions? *No*
- C.2.30 Does the medical device use an alarm system? *No*
- C.2.31 In what way(s) might the medical device be deliberately misused? *None*

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter	Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 4 of 9

- C.2.32 Does the medical device hold data critical to patient care? *No*
- C.2.33 Is the medical device intended to be mobile or portable? *Yes*
- C.2.34 Does the use of the medical device depend on essential performance? *No*

6.0 Risk Acceptability Determination

- 6.1 The criteria of risk acceptability will be determined with a Risk Priority Number (RPN) and on the concept of 'As Far As Possible' (AFAP) approach. Refer to the Risk Rating definition in §4.0.
- 6.2 There are two (2) components used in determining risk acceptability, which are Severity and Probability. Refer to the following tables:

Table 5.2a - Severity Ratings	
Index	Description
1	No patient effect
2	Minor injury that resolves without treatment
3	Moderate injury that may require treatment
4	Serious injury requiring medical intervention
5	Life threatening injury, disability, death

Table 5.2b – Probability Ratings	
Index	Description
1	Improbable - <0.001%
2	Remote – 0.001% to .01%
3	Occasional – 0.01% to 0.1%
4	Likely – 0.1% to 1%
5	Frequent - >1%

- 6.3 Risk acceptability will be determined per the following risk index:

Table 5.3 – Risk Rating Table						
Severity	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
	1	1	2	3	4	5
Probability						

7.0 Risk Assessment Tables

Table 6.0 –Kurin Blood Collection Device
--

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter										Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>										Rev.: A
											Page 5 of 9

#	Hazard Category	PH #	Potential Hazard	S	Potential Failure Mode	Risk Controls ¹	Verification	P ¹	R ¹	Risk Controls ²	P ²	R ²
1	Electromagnetic Energy	The medical device does not have any electromagnetic energy hazards.										
2	Radiation Energy	The medical device does not have any radiation energy hazards.										
3	Thermal Energy	The medical device does not have any thermal energy hazards.										
4	Mechanical Energy	The medical device does not have any mechanical energy hazards.										
5	Biological	5.1	Fever/ infection to patient	3	Device is not provided sterile/ insufficient sterilization process	Device will be terminally sterilized with validated method (Design)	Sterilization validation	1	3	N/A	N/A	N/A
					Device is reused	Warnings statement for single-use (Information)	IFU	2	6	N/A	N/A	N/A
					Inadequate sterile barrier system	Sterile barrier system will be designed for material that are commonly used for single-use medical device (Design)	Design verification - sterile barrier	1	3	N/A	N/A	N/A
						Warning will be incorporated for inspecting packaging prior to use as well not using the device if packaging is damaged (Information)	Stability	2	6	N/A	N/A	N/A
					Use beyond expiration date	Warning statement (Information)	IFU	2	6	N/A	N/A	N/A
		5.2	Fever/infection to clinician	3	Proper handling and disposal is not conducted during clinical application	Instruction and warning statement (Information)	IFU	2	6	N/A	N/A	N/A
6	Chemical	6.1	Allergic reaction	3	Materials (e.g., latex, rubber) that is known to cause allergic response	Material will be selected that do not cause allergic response (Design)	Design verification - biocompatibility	1	3	N/A	N/A	N/A
		6.2	False positives in the blood work evaluation – red blood cell damage	1	Material leach during clinical application	Material selection (Design)	Design verification - biocompatibility	1	1	N/A	N/A	N/A
					Excessive ethylene oxide residuals from sterilization process	Device will be released for distribution with acceptable residual levels (Design)	Sterilization validation	1	1	N/A	N/A	N/A

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter	Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 6 of 9

Table 6.0 –Kurin Blood Collection Device												
#	Hazard Category	PH #	Potential Hazard	S	Potential Failure Mode	Risk Controls ¹	Verification	P ¹	R ¹	Risk Controls ²	P ²	R ²
		6.3	Fever	3	Contacting material causes pyrogenic response	Material selection (Design)	Design verification - biocompatibility (i.e., Material mediated pyrogenicity)	1	3	N/A	N/A	N/A
		6.4	Procedure complication – device inoperative	3	Material degrades over usable life	Material selection (Design) Sterilization method selection (Design)	Stability evaluation	1	3	N/A	N/A	N/A
7	Biocompatibility	7.1	Cell Damage	3	Patient contact materials are not biocompatibility	Material selection (Design)	Design verification - biocompatibility	1	3	N/A	N/A	N/A
8	Function (Operational)	8.1	Procedure complication – device inoperative	2	Balloon Burst – Material or dimensional issues (wall thickness)	Material selection (Design) Dimensional Assessment (Design)	Design verification - Balloon Integrity (Resistance to Rupture)	1	2	N/A	N/A	N/A
				1	Balloon deflates or does not inflates – Valve malfunctions or leaks (wall thickness issue)	Valve selection (Design) Dimensional Assessment (Design)	Design verification - Balloon Volume Maintenance	1	1	N/A	N/A	N/A
				1	Balloon does not inflate – Valve malfunctions	Valve selection (Design)	Design verification	1	1	N/A	N/A	N/A
				2	Catheter breaks under normal tensile stresses - Material or dimensional issues	Valve selection (Design) Dimensional Assessment (Design)	Design verification - Inflated Balloon Response to Traction	1	2	N/A	N/A	N/A
				1	Connection issues – Syringe (inflation, irrigation); Collection bag (drainage)	Connection will be standardized to what is incorporated with marketed catheters (Design)	Design verification	1	1	N/A	N/A	N/A
		8.2	Procedure complication – downgraded performance	2	Poor drainage - Insufficient holes and lumen diameter, Lumen collapses (Materials or dimensional issues)	Material selection (Design) Dimensional Assessment (Design)	Design verification - Flow Rate though Drainage Lumen	1	2	N/A	N/A	N/A
				2	Poor irrigation - Insufficient holes and lumen diameter, Lumen collapses (Materials or dimensional issues)	Material selection (Design) Dimensional Assessment (Design)	Design verification	1	2	N/A	N/A	N/A

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter									Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>									Rev.: A
										Page 8 of 9

Table 6.0 –Kurin Blood Collection Device												
#	Hazard Category	PH #	Potential Hazard	S	Potential Failure Mode	Risk Controls ¹	Verification	P ¹	R ¹	Risk Controls ²	P ²	R ²
					Device used after expiration date or outside biocompatibility duration limits							
					Occurs during disposal process							
10	Labeling (Product Labeling)	10.1	Loss of traceability - Cannot identify the device or trace back to MFG	1	Incomplete or ambiguous instructions for use	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	N/A	N/A	N/A
		10.2	Device used outside specification (expiry/ single-use	1	Incomplete or ambiguous instructions for use	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	N/A	N/A	N/A
11	Operating Instructions (Product Labeling)	11.1	Procedural Complication /Patient discomfort	3	Incomplete or ambiguous instructions for use	IFU will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	3	N/A	N/A	N/A
12	Environmental	12.1	Device/ Sterile Barrier Failure /Patient health is compromised	3	Stored outside the prescribe environmental parameters	Warning statement (Information)	IFU	2	6	N/A	N/A	N/A
					Packaging is not adequate to protect the device through transportation	Packaging will be design to protect the device and it sterile barrier through transportation stresses (Design)	Design Verification (Distribution Evaluation)	1	3	N/A	N/A	N/A

8.0 Risk/ Benefit Assessment

8.1 The following table indicates the risk/benefit analysis for the residual risks that were greater than the risk acceptability limit after all applicable controls/mitigations were applied:

Table 7.1 – Risk/Benefit Assessment for Residual Risk above Acceptability Limit for Process Hazard Assessment										
Table No.	Process Description	Section No.	Potential Failure Mode	Potential Effect of Failure	Potential Cause of Failure	S	F ²	Risk/Benefit Analysis		Does the benefit outweigh the residual risk?
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		N/A

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter	Doc No.: RM-0019
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 9 of 9

8.2 There was no residual risk identified that was above the risk acceptability limit per §5.3.

9.0 Conclusion

9.1 Risks have been reduced as low as possible by the following methods:

- Design
- Appropriate testing to ensure design criteria were met
- Protective measures (critical-to-quality)
- Manufacturing and quality control measures

9.2 Based on a review of the risk-management process for the HDX Foley Catheter described herein, the risk management plan based on ISO 14971:2012 has been implemented and the individual, cumulative and overall residual risk have been reduced as low as possible regardless of the magnitude and the benefit of using this device will outweigh risks posed by the device during usage.

10.0 Change History

Revision	Change Description
A	Initial release

Pathway Innovation LLC.	Design Failure Mode Effects Analysis (FMEA) – Risk Management Report HDX Foley Catheter	Doc No.: RM-0020
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 1 of 8

1.0 Purpose

- 1.1 The purpose of this document is to identify the associated risk with the intended use of the HDX Foley Catheter as it relates to the design of the components and their interactions with each other.

2.0 Scope

- 2.1 The product numbers associated with the HDX Foley Catheter is indicated in the following table:

Table 2.1 – HDX Foley Catheter Products	
Part No.	Description
HDX-8000	14 Fr, 2-Way Foley Catheter
HDX-8001	16 Fr, 2-Way Foley Catheter
HDX-8002	18 Fr, 2-Way Foley Catheter

3.0 References

- 3.1 QP-004 – Risk Management

4.0 Risk Tables Definitions

- 4.1 Description – Describes the operation that is being evaluated for risk
- 4.2 Detection - The ability to screen out a hazard through verification activities (e.g., inspection).
- 4.2.1 Detection¹ (D¹) – The detection assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)
- 4.2.2 Detection² (D²) – The detection assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)
- 4.3 Verification – Evidence source of linkage that initial (i.e., Risk Controls¹) risk controls have implemented, verified, or validated.
- 4.4 Probability – Likelihood of the hazard occurring
- 4.4.1 Probability¹ (P¹) – The probability assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)
- 4.4.2 Probability² (P²) – The probability assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)
- 4.5 Potential Effect of Failure – All conceivable effects of the potential failure mode
- 4.6 Potential Cause of Failure – All conceivable causes of the potential failure mode
- 4.7 Potential Failure Mode (FM) – The potential consequences of the failure

Pathway Innovation LLC.	Design Failure Mode Effects Analysis (FMEA) – Risk Management Report	Doc No.: RM-0020
	HDX Foley Catheter	Rev.: A
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Page 2 of 8

4.8 Risk Controls - Process through which decisions are reached and protective measures are implemented for reducing risk to, or maintaining risk within specified levels.

4.8.1 Risk Controls¹ – Initial risk controls that are incorporated.

4.8.2 Risk Controls² – Secondary risk controls, also known as “Mitigation”.

4.9 Risk Rating – Residual risk determination based on the concept of ‘As Far As Possible’ (AFAP) approach. The AFAP approach provides three (3) possible outcomes, which are 1) the residual risk exceeds the risk acceptability (RED), 2) the residual risk is acceptable to the risk acceptability criteria (GREEN), and 3) the residual risk is between the two (2) states indicated and it can be an option to lower the risk to the lowest practical level without economic considerations (YELLOW).

4.9.1 Risk Rating¹ (R¹) – This calculation is post the implementation of initial risk controls (i.e., Risk Controls¹)

4.9.2 Risk Rating² (R²) – This calculation is post the implementation of the secondary risk controls (i.e., Risk Controls²)

4.10 Severity (S) – Measure of the possible consequences of a hazard.

5.0 Risk Acceptability Determination

5.1 The criteria of risk acceptability will be determined with a Risk Priority Number (RPN) and on the concept of ‘As Far As Possible (AFAP)’ approach. Refer to the Risk Rating definition in §4.0.

5.2 There are three (3) components used in determining risk acceptability, which are Severity, Probability, and Detection. Refer to the following tables:

Table 5.2a - Severity Ratings	
Index	Description
1	No patient effect
2	Minor injury that resolves without treatment
3	Moderate injury that may require treatment
4	Serious injury requiring medical intervention
5	Life threatening injury, disability, death

Table 5.2b – Probability Ratings	
Index	Description
1	Improbable - <0.001%
2	Remote – 0.001% to .01%
3	Occasional – 0.01% to 0.1%
4	Likely – 0.1% to 1%
5	Frequent - >1%

Index	Description
1	Very High – Current controls are almost certain to detect the failure mode.
2	High – High likelihood that current controls will detect the failure mode.
3	Moderate – Moderate likelihood that current controls will detect the failure mode.
4	Low – Low likelihood that current controls will detect the failure mode.
5	Remote – Remote likelihood that current controls will detect the failure mode.

5.3 Risk acceptability will be determined per the following risk index:

Severity	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
	1	1	2	3	4	5
=Square Root (Probability X Detection)						

Note: The purpose of using the square root of the probability and detection scores is to deemphasize these two (2) components, which places more emphasize on the severity component.

6.0 Risk Assessment Tables

Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
1	Catheter tube transfers urine from patient into drainage bag	Poor drainage / occluded lumen	2	Patient discomfort – device performance	Insufficient holes, lumen diameter, wall thickness	Design follows standard ASTM F623	TR-0072 (flow test)	1	2	2	4	N/A	N/A	N/A	N/A	N/A
		Catheter breaks	2	Procedural Complication – device failure	Wall thickness	Design follows standard ASTM F623	TR-0072 (inflated balloon response to traction)	1	2	2	4	N/A	N/A	N/A	N/A	N/A
					Performance degrades over time	Material Selection (Design)	Stability Evaluation TR-0073									

Pathway Innovation LLC.	Design Failure Mode Effects Analysis (FMEA) – Risk Management Report HDX Foley Catheter											Doc No.: RM-0020
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>											Rev.: A
												Page 6 of 8

Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
6	Information for Use (DFU) – Provides device information	Complication with device use/ intended purpose	3	Patient harm	Incomplete or ambiguous instructions for use	IFU will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	3	N/A	N/A	N/A	N/A	N/A
7	Inner Carton – Protects and orients the pouched devices	Packaging is damaged	3	Device/ sterile barrier damage	Inadequate design	Material/ card weight selection (Design)	Design verification (distribution evaluation) TR-0074	1	1	1	3	N/A	N/A	N/A	N/A	N/A
						Geometry is design to handle anticipate load (Design)										
8	Label, Inner Carton – Provides device information	Identification issues	1	Nuisance in preparation for the clinical application	Incomplete or ambiguous product information	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Device use outside specification	3	Patient harm	Inadequate disclosure of specifications/ limitation (e.g., single-use, expiry)	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	3	N/A	N/A	N/A	N/A	N/A
9	Shipper Case – Protects packaged	Packaging is damaged	3	Device/ sterile barrier damage	Inadequate design	Material/ card weight selection (Design)	Design verification (distribution	1	1	1	3	N/A	N/A	N/A	N/A	N/A

Pathway Innovation LLC.	Design Failure Mode Effects Analysis (FMEA) – Risk Management Report HDX Foley Catheter											Doc No.: RM-0020
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>											Rev.: A
												Page 7 of 8

Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
	devices					Geometry is design to handle anticipate load (Design)	evaluation) TR-0074									
10	Label, Shipper Case – Provides device information	Identification issues	1	Nuisance in preparation for the clinical application	Incomplete or ambiguous product information	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	1	N/A	N/A	N/A	N/A	N/A

7.0 Risk/ Benefit Assessment

7.1 The following table indicates the risk/benefit analysis for the residual risks that were greater than the risk acceptability limit after all applicable controls/mitigations were applied:

Table 7.1 – Risk/Benefit Assessment for Residual Risk above Acceptability Limit for Process Hazard Assessment									
Table No.	Process Description	Section No.	Potential Failure Mode	Potential Effect of Failure	Potential Cause of Failure	S	F ²	Risk/Benefit Analysis	Does the benefit outweigh the residual risk?
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

7.2 There was no residual risk identified that was above the risk acceptability limit per §5.3.

8.0 Conclusion

8.1 Risks have been reduced as low as possible by the following methods:

- Design the product with inherent safety
- Protective measures/ mechanisms to the product of the device.
- Provide information for safety

Pathway Innovation LLC.	Design Failure Mode Effects Analysis (FMEA) – Risk Management Report HDX Foley Catheter	Doc No.: RM-0020
	<i>This document contains confidential, proprietary information of Pathway Innovation LLC and shall not be copied or reproduced without prior written permission there from. If any portion of this document is in hardcopy form, it must be considered and treated as an uncontrolled. Users of this document must verify that they are using the most current revision of the controlled document before performing work.</i>	Rev.: A
		Page 8 of 8

Based on a review of the risk-management process for the HDX Foley Catheter described herein, the risk management plan based on ISO 14971:2012 has been implemented and the individual, cumulative and overall residual risk have been reduced as low as possible regardless of the magnitude and the benefit of using this device will outweigh risks posed by the device during usage.

9.0 Change History

Revision	Change Description
A	Initial release